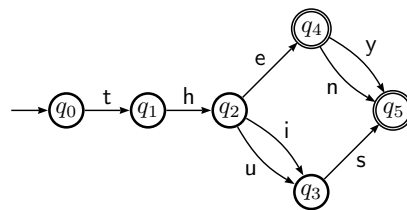


Final examination of Information Retrieval

Christian Rinderknecht

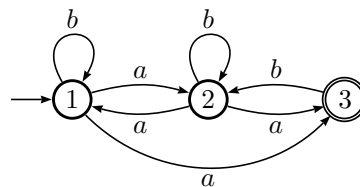
15 December 2005

1. Consider the following trie, which, given an input string, will either be stuck or accept one of the English words “the”, “then”, “they”, “this” or “thus” at the *beginning* of the string.



Modify the trie such that it recognises the words *anywhere* in the string.

2. Consider the following NFA $\mathcal{N}_1$



and prove that the word *aaba* is accepted or prove the contrary.

3. Extend the subset construction to cope with NFA with *several* initial states and apply this extended version to

